

## MA'AN, Jordan

WMO: 403100

Lat: 30.167N

Lon: 35.783E

Elev: 1069

StdP: 89.12

Time Zone: 2.00 (JOR)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -0.5       | 0.9 | -11.1                         | 1.6 | 11.0 | -8.9 | 2.0 | 12.5 | 18.9                  | 5.2  | 15.6 | 4.4  | 1.4                   | 280  | 0.417 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.8                   | 36.6            | 19.9 | 35.4 | 19.4 | 34.2 | 18.8 | 21.9                | 33.5 | 21.1 | 32.2 | 20.3 | 31.0 | 2.9                  | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.7                            | 15.5 | 25.4 | 17.8 | 14.5 | 25.6 | 16.8 | 13.6 | 25.3 | 69.8          | 33.2 | 66.2 | 32.1 | 63.4 | 30.7 | 26.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| 13.0              | 10.3 | 8.7 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |     | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |     | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 18.2   | 7.8   | 9.7   | 13.7  | 17.2  | 22.3  | 25.2  | 26.9  | 27.0  | 24.5  | 20.3  | 13.9  | 9.4  |
|                                                                                  | DBStd       | 7.53   | 3.17  | 3.37  | 4.02  | 4.21  | 3.41  | 2.45  | 2.38  | 2.14  | 2.32  | 3.24  | 3.06  | 3.50 |
|                                                                                  | HDD10.0     | 196    | 86    | 42    | 11    | 2     | 0     | 0     | 0     | 0     | 0     | 4     | 52    |      |
|                                                                                  | HDD18.3     | 1230   | 328   | 243   | 152   | 72    | 7     | 0     | 0     | 0     | 18    | 136   | 276   |      |
|                                                                                  | CDD10.0     | 3188   | 16    | 32    | 127   | 217   | 381   | 457   | 523   | 527   | 434   | 319   | 122   | 34   |
|                                                                                  | CDD18.3     | 1179   | 0     | 0     | 9     | 36    | 128   | 207   | 265   | 268   | 184   | 78    | 3     | 1    |
|                                                                                  | CDH23.3     | 13404  | 0     | 5     | 107   | 416   | 1444  | 2409  | 3211  | 3152  | 1889  | 732   | 34    | 7    |
|                                                                                  | CDH26.7     | 6346   | 0     | 0     | 24    | 118   | 590   | 1181  | 1677  | 1674  | 848   | 231   | 2     | 1    |
| Wind                                                                             | WSAvg       | 2.7    | 2.9   | 3.0   | 3.7   | 3.6   | 3.2   | 2.8   | 2.6   | 2.5   | 2.3   | 1.9   | 2.0   | 2.1  |
| Precipitation                                                                    | PrecAvg     | 40     | 9     | 7     | 5     | 3     | 2     | 0     | 0     | 0     | 5     | 3     | 6     |      |
|                                                                                  | PrecMax     | 95     | 46    | 46    | 28    | 28    | 26    | 4     | 0     | 5     | 53    | 22    | 41    |      |
|                                                                                  | PrecMin     | 5      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |      |
|                                                                                  | PrecStd     | 22     | 10    | 9     | 6     | 5     | 5     | 1     | 0     | 1     | 10    | 5     | 8     |      |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 20.8  | 24.2  | 29.7  | 32.6  | 35.9  | 36.6  | 38.7  | 38.2  | 36.4  | 32.8  | 26.7  | 23.7 |
|                                                                                  |             | MCWB   | 10.1  | 12.0  | 13.7  | 16.1  | 18.2  | 18.6  | 21.2  | 21.8  | 18.4  | 16.7  | 14.1  | 11.8 |
|                                                                                  | 2%          | DB     | 18.4  | 21.2  | 26.9  | 30.0  | 33.5  | 35.2  | 36.7  | 36.4  | 34.0  | 31.0  | 24.1  | 20.7 |
|                                                                                  |             | MCWB   | 9.2   | 9.9   | 12.8  | 14.4  | 17.1  | 18.5  | 20.3  | 20.4  | 18.0  | 16.0  | 12.9  | 10.4 |
|                                                                                  | 5%          | DB     | 16.4  | 18.6  | 24.0  | 27.8  | 31.9  | 34.0  | 35.2  | 35.1  | 32.6  | 29.4  | 22.1  | 18.0 |
|                                                                                  |             | MCWB   | 8.5   | 8.8   | 11.6  | 13.6  | 16.7  | 18.0  | 19.7  | 19.9  | 17.5  | 15.7  | 12.2  | 10.0 |
|                                                                                  | 10%         | DB     | 14.3  | 16.5  | 21.7  | 25.7  | 30.2  | 32.7  | 33.6  | 33.7  | 31.3  | 27.6  | 20.1  | 15.6 |
|                                                                                  |             | MCWB   | 7.6   | 8.1   | 10.7  | 12.9  | 16.0  | 17.2  | 19.0  | 19.1  | 17.2  | 15.3  | 11.6  | 8.7  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 12.1  | 12.7  | 15.4  | 18.2  | 20.0  | 21.5  | 23.0  | 23.3  | 21.5  | 19.1  | 16.5  | 13.8 |
|                                                                                  |             | MCDB   | 17.5  | 19.5  | 25.2  | 29.5  | 31.1  | 33.5  | 33.7  | 34.9  | 31.8  | 27.6  | 19.7  | 20.0 |
|                                                                                  | 2%          | WB     | 10.4  | 11.0  | 13.7  | 16.0  | 18.7  | 20.3  | 21.6  | 22.3  | 20.0  | 17.9  | 14.5  | 11.9 |
|                                                                                  |             | MCDB   | 16.5  | 19.1  | 24.1  | 26.1  | 30.9  | 32.8  | 33.8  | 33.5  | 29.5  | 26.6  | 21.0  | 18.1 |
|                                                                                  | 5%          | WB     | 9.1   | 9.8   | 12.5  | 14.6  | 17.8  | 19.3  | 20.7  | 21.4  | 19.1  | 17.0  | 13.5  | 10.5 |
|                                                                                  |             | MCDB   | 15.3  | 16.6  | 22.7  | 25.8  | 29.8  | 31.5  | 32.3  | 32.2  | 28.5  | 25.3  | 19.2  | 16.8 |
|                                                                                  | 10%         | WB     | 7.8   | 8.8   | 11.2  | 13.5  | 16.7  | 18.3  | 20.0  | 20.6  | 18.3  | 16.2  | 12.7  | 9.3  |
|                                                                                  |             | MCDB   | 13.4  | 15.4  | 20.6  | 24.4  | 28.6  | 30.2  | 31.5  | 30.9  | 27.9  | 24.9  | 18.3  | 14.6 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.3  | 11.1  | 12.2  | 12.9  | 13.8  | 14.5  | 13.8  | 14.4  | 14.1  | 13.1  | 11.5  | 11.0 |
|                                                                                  |             | MCDBR  | 13.2  | 14.6  | 14.8  | 15.1  | 14.9  | 15.6  | 14.8  | 15.1  | 14.9  | 14.6  | 14.1  | 14.0 |
|                                                                                  |             | MCWBR  | 7.5   | 7.2   | 6.9   | 7.1   | 6.3   | 6.1   | 6.4   | 6.1   | 6.1   | 6.0   | 7.0   | 7.5  |
|                                                                                  | 5% WB       | MCDBR  | 12.4  | 12.7  | 13.5  | 14.0  | 14.3  | 15.3  | 14.1  | 14.8  | 14.4  | 13.3  | 11.8  | 12.3 |
|                                                                                  |             | MCWBR  | 7.6   | 7.2   | 7.1   | 7.3   | 7.0   | 6.9   | 6.6   | 6.3   | 6.6   | 6.1   | 6.5   | 7.3  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.306  | 0.337 | 0.386 | 0.429 | 0.431 | 0.363 | 0.365 | 0.375 | 0.370 | 0.380 | 0.329 | 0.302 |      |
|                                                                                  | taud        | 2.368  | 2.258 | 2.081 | 1.971 | 1.998 | 2.235 | 2.244 | 2.225 | 2.232 | 2.189 | 2.332 | 2.396 |      |
|                                                                                  | Ebn at Noon | 927    | 923   | 895   | 866   | 863   | 919   | 915   | 905   | 899   | 862   | 889   | 912   |      |
|                                                                                  | Edh at Noon | 101    | 123   | 158   | 183   | 179   | 141   | 139   | 140   | 134   | 131   | 104   | 93    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 3.66   | 4.61  | 5.95  | 6.96  | 7.76  | 8.46  | 8.15  | 7.53  | 6.56  | 5.20  | 4.07  | 3.44  |      |
|                                                                                  | RadStd      | 0.16   | 0.20  | 0.21  | 0.17  | 0.29  | 0.10  | 0.15  | 0.12  | 0.18  | 0.18  | 0.15  | 0.13  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (12 neighbors) | +0.44 | N/A     | -1.65  | N/A     | N/A   | N/A   | N/A         | N/A     | +151    | +105    |

Nomenclature: See separate page